

Task 1: Configure SSH with Key-Based Authentication

Install OpenSSH Server

- `sudo dnf install openssh-server -y`

Start and Enable SSH

- `sudo systemctl enable --now sshd`

Generate SSH Key Pair

- `ssh-keygen -t rsa -b 4096`

Copy Public Key

- `cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys`
- `chmod 700 ~/.ssh`
- `chmod 600 ~/.ssh/authorized_keys`

Disable Password Authentication

Edit:

- `sudo vi /etc/ssh/sshd_config`

Change:

- `PasswordAuthentication no`
- `PubkeyAuthentication yes`
- `PermitRootLogin no`

Restart SSH:

- `sudo systemctl restart sshd`

Verify:

- `ssh -v localhost`

Task 2: Configure Firewall

Start Firewall:

- `sudo systemctl enable --now firewalld`

Allow SSH:

- `sudo firewall-cmd --permanent --add-service=ssh`

Allow HTTP:

- `sudo firewall-cmd --permanent --add-service=http`

Reload Firewall:

- `sudo firewall-cmd --reload`

Verify:

- `sudo firewall-cmd --list-all`

Task 3: Create Backup Script

Create directory:

- `mkdir -p ~/scripts`

Create script:

- `vim ~/scripts/automate_backup.sh`

Paste:

```
#!/bin/bash
DATE=$(date +%F_%H-%M)
DEST="/backups/home_backup_$(date)"
mkdir -p "$DEST"
cp -r /home/* "$DEST"
echo "Backup completed on $(date)"
```

Make executable:

- `chmod +x ~/scripts/automate_backup.sh`

Run manually:

- `~/scripts/automate_backup.sh`

Schedule Backup Using Cron

Open crontab:

- `crontab -e`

Add:

- `0 2 * * * /home/student/scripts/automate_backup.sh >> /home/student/backup.log 2>&1`

Verify:

- `crontab -l`

Check log:

- `tail -5 ~/backup.log`

Task 4: Configure SELinux

Check status:

- `getenforce`

Detailed status:

- sestatus

Verify:

- getenforce